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An automatic wash basin tap is quite helpful in maintaining hygiene as it does not need to be touched for water flow. It is especially useful in public places like restaurants and hospitals where a large number of people use the wash basins. Since the water stops flowing after use, it also prevents water wastage.

This circuit lets the water flow from the tap automatically when a person comes in front of the wash basin to wash hands. The water stops flowing as soon as the person tries to leave and removes his hands away from the wash basin. The author's prototype is shown in Fig. 1.

Circuit diagram for the automatic

wash basin tap is shown in Fig. 2. The circuit comprises an Arduino board (BOARD1), ultrasonic sensor HC-SR04, 12V normally closed solenoid valve, and a few other components. The wash basin's solenoid valve (to let water flow from the tap to which it is connected) opens automatically when a user's hands are detected just 3 to 5cm away by the ultrasonic sensor HC-SR04.

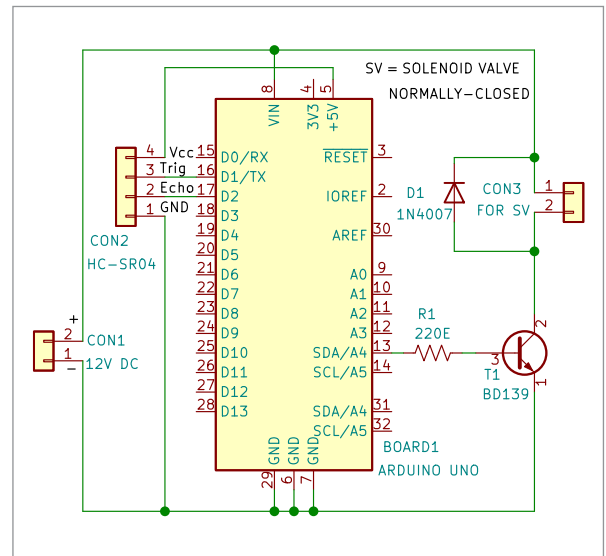


Fig. 2: Circuit diagram

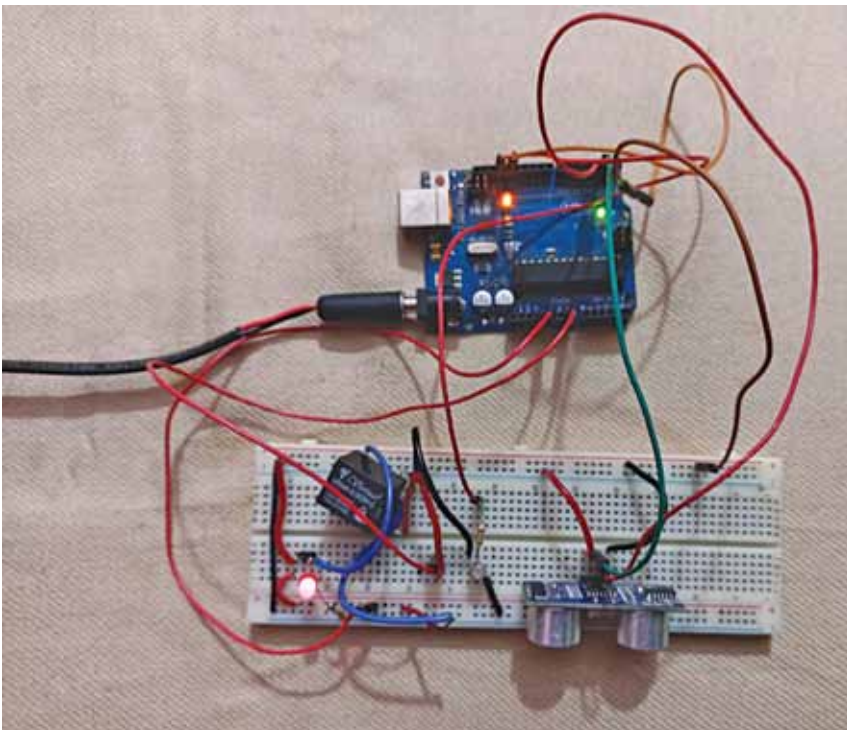


Fig. 1: Author's prototype on a breadboard



Fig. 3: Ultrasonic sensor HC-SR04

Fig. 3 shows the front side of ultrasonic sensor HC-SR04, which is the heart of this circuit. The sensor can be easily interfaced with Arduino Uno. When a hand is less than 5cm away from the sensor, it reads and stores the value (distance), and the relay is turned on to perform switching action for opening the solenoid valve to let water flow.

When the hand moves away